

COMP 324 Network Security

Module 3 Lab 2: Permission Manipulation

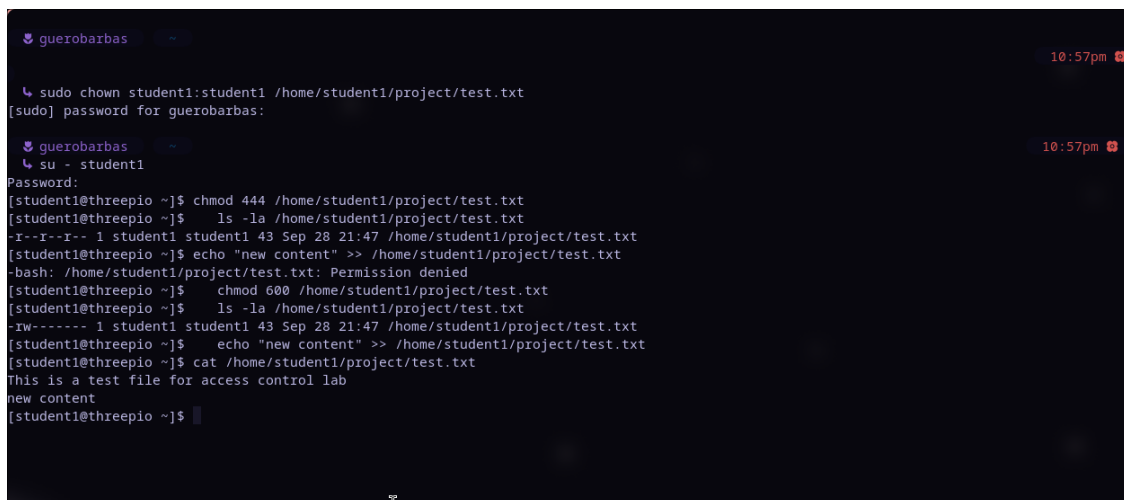
Lab Report

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Date: September 28, 2025

Step 1: Changing File Permissions



```
gueroabaras ~  
↳ sudo chown student1:student1 /home/student1/project/test.txt  
[sudo] password for gueroabaras:  
gueroabaras ~  
↳ su - student1  
Password:  
[student1@threepio ~]$ chmod 444 /home/student1/project/test.txt  
[student1@threepio ~]$ ls -la /home/student1/project/test.txt  
-r--r--r-- 1 student1 student1 43 Sep 28 21:47 /home/student1/project/test.txt  
[student1@threepio ~]$ echo "new content" >> /home/student1/project/test.txt  
-bash: /home/student1/project/test.txt: Permission denied  
[student1@threepio ~]$ chmod 600 /home/student1/project/test.txt  
[student1@threepio ~]$ ls -la /home/student1/project/test.txt  
-rw----- 1 student1 student1 43 Sep 28 21:47 /home/student1/project/test.txt  
[student1@threepio ~]$ echo "new content" >> /home/student1/project/test.txt  
[student1@threepio ~]$ cat /home/student1/project/test.txt  
This is a test file for access control lab  
new content  
[student1@threepio ~]$
```

Explanation: The file was first returned to `student1` ownership using `chown`. `chmod 444` set the file to read-only for all users, append failed, verifying read-only enforcement. Next, `chmod 600` gave read/write access only to the owner. Verification with `ls -la` showed `-rw----`, and appending content succeeded, confirming the permission change.

Step 2: Special Permissions

```
[student1@threepio ~]$ echo '#!/bin/bash'
echo "Current user: $(whoami)"
echo "Effective user: $(id -un)" > /home/student1/project/test_script.sh
[student1@threepio ~]$ chmod +x /home/student1/project/test_script.sh
[student1@threepio ~]$ exit
logout

guero@barbas ~$ sudo chmod 4755 /home/student1/project/test_script.sh 11:03pm
[sudo] password for guero@barbas:
guero@barbas ~$ su - student1 11:03pm
Password:
[student1@threepio ~]$ ls -la /home/student1/project/test_script.sh
-rwsr-xr-x 1 student1 student1 76 Sep 28 23:02 /home/student1/project/test_script.sh
[student1@threepio ~]$ /home/student1/project/test_script.sh
Current user: student1
Effective user: student1
[student1@threepio ~]$ exit
logout

guero@barbas ~$ sudo chmod 2755 /home/student1/project/ 11:05pm
guero@barbas ~$ sudo chmod 1755 /home/student1/project/ 11:05pm
guero@barbas ~$ 11:05pm
```

Explanation:

`chmod +x` added execute permission. `chmod 4755` set SUID, changing owner execute to `s`. `chmod 2755` set SGID, changing group execute to `s`. `chmod 1755` set the Sticky Bit, adding `t` in the others' field. **Observations:**

- The SUID bit replaces the owner's execute bit with `s`, making programs run with file owner privileges.
- SGID on a directory ensures all new files inherit the directory's group
- The Sticky Bit on a directory prevents users from deleting others' files

Step 3: Access Control Lists (ACLs)

```
guero@barbas ~$ sudo pacman -S acl
[sudo] password for guero@barbas:
warning: acl-2.3.2-1 is up to date -- reinstalling
resolving dependencies...
looking for conflicting packages...

Packages (1) acl-2.3.2-1

Total Installed Size: 0.32 MiB
Net Upgrade Size: 0.00 MiB

:: Proceed with installation? [Y/n] y
(1/1) checking keys in keyring [#####] 100%
(1/1) checking package integrity [#####] 100%
(1/1) loading package files [#####] 100%
(1/1) checking for file conflicts [#####] 100%
(1/1) checking available disk space [#####] 100%
:: Processing package changes...
(1/1) reinstalling acl [#####] 100%
:: Running post-transaction hooks...
(1/1) Arming ConditionNeedsUpdate...

guero@barbas ~$ sudo setfacl -m u:student1:r /home/student1/project/test.txt

guero@barbas ~$ sudo setfacl -m g:developers:w /home/student1/project/test.txt

guero@barbas ~$ su - student1
Password:
[student1@threepio ~]$ getfacl /home/student1/project/test.txt
getfacl: Removing leading '/' from absolute path names
# file: home/student1/project/test.txt
# owner: student1
# group: student1
user::rw-
user:student1:r--
group::---
group:developers:-w-
mask::rw-
other::---

[student1@threepio ~]$ exit
logout

guero@barbas ~$ sudo setfacl -b /home/student1/project/test.txt

guero@barbas ~$
```

Explanation:

A user ACL entry was added with `setfacl -m u:student1:r`, with read access, and a group ACL entry with `setfacl -m g:developers:w`, granting write access. `getfacl` showed the additional ACL entries alongside the standard owner/group/other fields. ACLs were removed with `setfacl -b`, restoring default permissions.

Output:

```
# file: home/student1/project/test.txt
# owner: student1
# group: student1
user::rw-
user:student1:r--
group::---
group:developers:w-
```

```
mask::rw-  
other::---
```

This confirms that ACLs can add specific user and group permissions independent of the standard permission bits.

Conclusion

This lab showed me how changing permissions and ownership directly controls what users can do with files. I learned how bits like SUID, SGID, and the Sticky Bit change behavior, and how ACLs give more fine-grained control. Overall, it made permissions feel a lot clearer and less confusing.